Philipp Benjamin Aretz
Center for Theoretical Physics, Massachusetts Institute of Technology
Cambridge, MA, USA

Research interests

Effective field theory and precision QCD; factorization and resummation for collider observables; medium effects in heavy-ion collisions; mathematical structure of quantum field theory.

Education

PhD in Physics, Massachusetts Institute of Technology, Cambridge, MA, USA 2024–present

MASt in Mathematics (Part III), University of Cambridge, Cambridge, UK 2023–2024
Pass with Distinction, 92% — ranked 9th of 294

BSc in Physics, Heidelberg University, Heidelberg, Germany 2020–2023
1.0 (best possible)
Thesis: *The Keldysh Functional Integral and Bounds on Truncated Expectation Values* (advisor: Prof. Manfred Salmhofer)

Abitur, Goethe-Schule Bochum, Bochum, Germany 2020
1.0

Research positions

Research Assistant, Heidelberg University, Heidelberg, Germany 2022–2023
CRC/TRR 191 Symplectic Structures in Geometry, Algebra and Dynamics
Differential delay equations: periodic delay orbits and smooth orbit families. Magnetic billiards: KAM-based extension near convex boundaries.

Tutor, Heidelberg University, Heidelberg, Germany 2021–2022

Publications and preprints

Published

P. B. Aretz, M. Salmhofer, *A rigorous Keldysh functional integral for fermions*, J.Statist.Phys. 193 (2026) 28, arXiv:2508.01787 [math-ph].

P. Albers, **P. Aretz**, I. Seifert, *Families of periodic delay orbits*, J. Differential Equations 410 (2024) 228-250, arXiv:2304.07550 [math].

Preprints

P. B. Aretz, K. Lee, S. T. Schindler, I. W. Stewart, *Factorization of elastic, single, and double diffractive pp scattering*, arXiv:2607.07788 [hep-ph].

Contributed conference talks

Diffraction at the LHC, ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna 2026

Generalised Detectors in Heavy Ion Collisions, ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna 2026

Diffraction Across Colliders, TASI - Exploring Fundamental Physics from Colliders to the Cosmos, Boulder 2026

Diffraction Across Colliders, SCET 2026, Seoul, South Korea 2026

Internal seminars

Generalised Detector Observables in the Heavy Ion Environment, MIT EFT Group Meeting (2025)

Resurgence and Renormalons in QCD, MIT EFT Group Meeting (2025)

Generalised Detector Observables in the Heavy Ion Environment, MIT Heavy Ions Group Meeting (2025)

Conferences and schools attended

ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Erwin Schrödinger Institute - Vienna (Poster) 2026

TASI - Exploring Fundamental Physics from Colliders to the Cosmos, Boulder, CO, USA 2026

SCET 2026, Seoul, South Korea (Talk) 2026

GGI School on Theory of Fundamental Interactions, Florence, Italy 2026

SCET 2025, Tucson, AZ, USA 2025

Awards and funding

Tom Frank Fellowship, Massachusetts Institute of Technology 2024

Jennings Prize, University of Cambridge 2024

DAAD Scholarship, Deutscher Akademischer Austauschdienst 2023

Wolfson College Scholarship, Wolfson College, Cambridge 2023

e-fellows.net Scholarship

Teaching

Classical Mechanics 8.01(TA), MIT Fall

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Service and leadership

Treasurer, Edgerton House, MIT 2025–2026

President MIT Rowing Club, MIT Rowing Club 2026–present

Organiser, First Year Seminar, Massachusetts Institute of Technology 2024–2025

Buddy for first-year students, Heidelberg University 2021–2022

Technical skills

Python (NumPy, pandas, TensorFlow); C++ (ROOT, Pythia); Mathematica; MATLAB; LaTeX.

Languages

German (native); English (C2, Cambridge Grade A, TOEFL 117/120); French (basic); Spanish (basic); Latin (Latinum).

References available on request.